

Techno-economic analysis of using ammonia as an energy carrier for renewable energy conversion and storage

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ABSTRACT

Using ammonia as an energy carrier for renewable energy conversion and storage via electricity-fuel-electricity approach has attracted ever-increasing attention during the last few decades. However, the technical feasibility and economic benefits of electricity-ammonia-electricity route remain to be investigated. Meanwhile, the integration and matching of various energy conversion and storage technologies deserve to be optimized towards high energy utilization efficiency. Therefore, in this study, a comprehensive techno-economic analysis of using ammonia as an energy carrier via the electricity-ammonia-electricity approach has been carried out. A renewable energy conversion and storage roadmap consisting of five different routes is proposed and analyzed. Besides, the effects of the system scale as well as the maturity of the technology on the system cost and efficiency are investigated. The estimated results demonstrate that the current achieved round-trip efficiency of electricity-ammonia-electricity ranges from 18.71 to 33.03 %. Besides, the cost of proton exchange membrane fuel cells-based electricity-ammonia-electricity system is the lowest, at about 0.06 USD/kWh. Furthermore, scaling up the system from 1 MW to 100 MW reduces the system cost by about 10 %. Additionally, from 2023 to 2050, with the development of technology, the cost of electricity-ammonia-electricity will decrease by around 47.7–59.6 %, and the efficiency will increase by approximately 17.3–40.7 %.

1. Introduction

Renewable energy, which can generate electricity from abundant clean and renewable resources such as solar and wind, has therefore attracted ever-increasing attention during the last few decades [1]. However, the intermittent nature of renewable energy makes it difficult to directly integrate renewable electricity into the grid, limiting its large-scale application [2]. Therefore, developing advanced technologies for storing and converting unstable renewable energy is necessary. Following this, an emerging energy storage technology is to develop an energy carrier, which can convert intermittent electricity from

renewable resources to stable electricity for end-users via an electricity-fuel-electricity approach [3]. Hydrogen (H₂), one of the cleanest energy carriers in the world, can be generated via the electrolysis of water (energy efficiency is around 70–80 %) [4]. Meanwhile, hydrogen can also be compressed in a liquid state and then used as a fuel to generate electricity in a fuel cell (~60 %) [5]. The round-trip efficiency of the electricity-fuel-electricity conversion route arrives at 40–50 % [6]. However, hydrogen, as the lightest gas in the world, is extremely hard to compress and liquefy (–253 °C under ambient pressure), which therefore limits its application at large scale [7]. As a result, an alternative energy carrier, ammonia (NH₃), which can be liquefied at

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–33 °C at ambient pressures, has attracted ever-increasing attention [8]. Besides, ammonia has a high hydrogen content (17.65 wt %) [9]. Furthermore, the infrastructure for ammonia storage and transportation is well-established and developed. Following these considerations, ammonia offers great potential for being an energy carrier for renewable energy conversion and storage.

Using ammonia as an energy carrier via an electricity-fuel-electricity approach includes two connectable processes: ammonia production and electricity generation [10]. Currently, large-scale ammonia production mainly relies on the traditional Haber Bosch process (producing grey ammonia at million tons scale with significant CO₂ emissions), improved Haber Bosch process (integrating CO₂ capture and storage and producing blue ammonia at million tons scale), and upgraded Haber Bosch process (coupling water electrolysis and producing green ammonia at million tons scale) [11]. To investigate the feasibility of the upgraded Haber Bosch process, Richard et al. [12] analyzed the production costs of green ammonia in accordance with 534 regions in 70 countries worldwide. Their analysis demonstrated that the achievable production cost of green ammonia in 2020 is about \$473/tonne, while this cost can be reduced to \$310/tonne by 2030. Besides, they also proved that the cost of the electrolyzer for hydrogen production, including capital expenditures and operating costs, is the largest component of the green ammonia production process. In addition to the Haber Bosch process, ammonia could also be produced via electrochemical methods such as electrocatalytic nitrogen reduction or nitrate reduction, which has been proposed and developed in recent years [13]. Advantages of the electrochemical method include zero emissions, mild working conditions, and size flexibility. Meanwhile, this technology allows for the direct conversion of abundant and cost-effective natural resources, such as nitrogen and water, into high-value chemicals [14]. However, this technology is still at the initial research exploration stage, continuous efforts should be carried out before widespread commercialization becomes possible.

After being produced, ammonia can be stored, transported, and distributed to generate electricity for end-user applications [15]. Regarding transportation, ammonia can be shipped via pipelines, shipping, or trucks, etc [16]. As the infrastructure of ammonia shipping is globally well-established, the storage and transportation costs would be much cheaper than that of hydrogen. Once the ammonia is prepared, electricity can be generated on-site via various energy conversion technologies. For example, Dincer et al. [17] explored the feasibility of using ammonia as a fuel in the internal combustion engine (ICE). Their findings indicate that the application of ammonia reduces carbon emissions compared to the use of conventional fuels. However, due to the low energy density and calorific value of ammonia, it needs to be mixed with other fuels [18], such as hydrogen. Additionally, the application of ammonia in conventional generators may generate new pollutants, such as nitrogen oxides, due to insufficient combustion [19]. In addition to the ICE application, ammonia can also be directly applied in fuel cells [20], such as solid oxide fuel cells (SOFC) and anion exchange membrane fuel cells (AEMFC), which demonstrate the capability to convert the chemical energy stored in the ammonia to the electrical energy. Zhang et al. [21] investigated the economic benefits and technical feasibility of using an ammonia fuel cell system to power railway freight transport. Their results showed that the ammonia-fed solid oxide fuel cell system had the best power generation efficiency of 56 %. In addition to the above-mentioned direct applications, ammonia can also be indirectly utilized for electricity generation [22]. For example, ammonia can be first cracked into hydrogen, and then the purified hydrogen can be fed into proton exchange membrane fuel cells (PEMFC) and SOFC to generate electricity [23]. In this process, ammonia is treated as a hydrogen carrier, whose feasibility has been verified by Lin et al. work [24]. They developed an advanced NH₃ cracking catalyst (Ru/C) that can reduce the reaction temperature by at least 300 °C compared to conventional catalysts. In addition, an indirect ammonia PEMFC (IA-PEMFC) system was analyzed and the results of the study

showed that the optimized system had a lower cost of power generation (\$0.13/kWh) compared to conventional systems using gasoline (\$0.18/kWh) and direct hydrogen fuel (\$0.5/kWh).

While previous studies have conducted economic analyses on both ammonia synthesis and ammonia-based power generation processes, the majority of this work has only focused on a single segment or partial conversion steps within the ammonia-as-energy-carrier (electricity-ammonia-electricity) chain. Furthermore, technological advancements have rendered earlier analyses inadequate for addressing future energy demands, such as requirements for higher energy efficiency and larger-scale energy application scenarios. To address these problems, in this study, we conduct a comprehensive techno-economic analysis (TEA) of ammonia as an energy carrier for renewable energy conversion and storage, focusing on the complete electricity-ammonia-electricity cycle. The core objective was to evaluate the technical feasibility and economic viability of integrating renewable-powered water electrolysis (AWE, PEMWE) with the Haber-Bosch process for ammonia production, paired with five distinct ammonia-to-electricity pathways: internal combustion engines (ICE), solid oxide fuel cells (SOFC), direct low-temperature ammonia fuel cells (AEMFC), and hydrogen-fueled proton exchange membrane fuel cells (PEMFC) or SOFC via ammonia cracking. The main novelty lies in its holistic comparison of these pathways—encompassing current performance, 2023–2050 projections, and scale-up effects—providing a unified framework to assess ammonia's role as a cross-sectoral energy vector, which has been lacking in fragmented prior analyses.

2. Methodology

2.1. Electricity-ammonia-electricity system configurations

The roadmap of using ammonia as an energy carrier for renewable energy conversion and storage via electricity-ammonia-electricity approach is shown in Fig. 1. It consists of two connectable processes, including ammonia production (electricity to ammonia) and electricity generation (ammonia to electricity). In this study, five different energy conversion routes will be introduced, which are named by the final application technologies, such as ICE-based electricity-ammonia-electricity route and SOFC-based electricity-ammonia-electricity route.

2.1.1. Electricity to ammonia

Electricity to ammonia currently can be realized via the improved Haber-Bosch process, which couples the green hydrogen production module via water electrolysis with the traditional Haber-Bosch process, it allows a massive ammonia production rate of 400 tonnes per day per ton of catalyst at standard pressures (150–200 atm) and temperatures (400–500 °C) [25]. The technical route of the improved Haber Bosch process is shown in Fig. 1. Firstly, electricity from renewable sources (e. g. solar or wind) or directly from the grid is connected to a water electrolysis system to produce green hydrogen. Thereafter, the green hydrogen, together with nitrogen separated from air, is fed into a reactor to produce green ammonia. Regarding the electricity generation (temporarily not considering the average price of regions), according to the report published by the International Renewable Energy Agency (IRENA), the global cost of onshore wind and solar power generation in 2022 are \$0.033/kWh and \$0.049/kWh, respectively, which are lower than fossil fuel (\$0.101/kWh) [26]. Therefore, in this study, the sources of renewable electricity are mainly considering generated from onshore wind and solar. In terms of the production of green hydrogen, the technologies that have been commercialized include alkaline water electrolysis (AWE) and proton exchange membrane water electrolysis (PEMWE). In this study, both technologies for hydrogen production will be considered. As for the cost of ammonia production, the Haber-Bosch process will be analyzed. In summary, the system cost and efficiency of electricity to ammonia are mainly affected by renewable electricity generation, hydrogen production, and ammonia production, which will

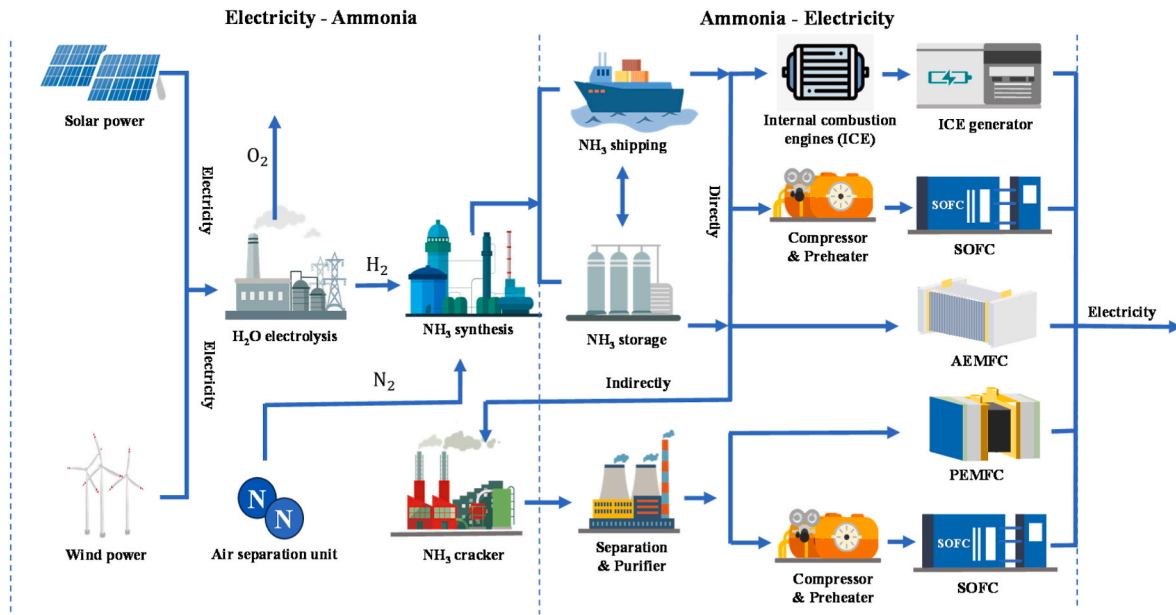


Fig. 1. Renewable energy conversion and storage via electricity-ammonia-electricity routes.

be introduced in the later section.

2.1.2. Ammonia to electricity

After being produced, ammonia needs to be transported from the place of production to the application terminal, which then was used to generate stable electricity for applications. Currently, the transportation of ammonia includes road transport, shipping, pipeline transport, etc. According to a report released by IRENA in 2022, a total of 25–30 Mt of ammonia is transported globally each year, of which about 18–20 Mt is transported by ship [26]. After being shipped to the application terminal, ammonia can be used for electricity generation via various energy conversion technologies. One typical application is to be used in ICE [27].

Ammonia-fueled ICEs convert the chemical energy of ammonia into mechanical energy through combustion, which is then converted into electricity via a generator. The working principle of the ammonia-fed ICE is injecting the ammonia into the engine cylinder, mixed with air, and ignited (often with a pilot fuel like diesel or spark ignition due to ammonia's high auto-ignition temperature of $\sim 651^\circ\text{C}$). The combustion produces high-pressure gas that drives the piston, rotating the crankshaft to generate mechanical power. The full-load efficiency ranges from

20 to 35 %, dropping significantly to 15–25 % at 50 % load (Table 1) due to unstable combustion and increased heat losses. In addition to ICE application, ammonia can also be directly used as a fuel in fuel cells, including ammonia-fed SOFC (600–800 $^\circ\text{C}$) [28] and ammonia-fed AEMFC (25–100 $^\circ\text{C}$) [29]. Regarding ammonia-fed SOFC, it directly utilizes ammonia as a fuel in a high-temperature electrochemical reaction, bypassing the need for cracking. The working principle is that ammonia reacts with oxygen (from air) at the anode of the SOFC, where it undergoes oxidation ($\text{NH}_3 + 3\text{O}^{2-} \rightarrow \text{N}_2 + 3\text{H}_2\text{O} + 6\text{e}^-$) at high temperatures [30]. Electrons flow through an external circuit, generating electricity. Full-load efficiency is 40–60 %, with more stable part-load performance (35–55 % at 50 % load) due to robust electrochemical reactions at 600–1000 $^\circ\text{C}$, enabling internal ammonia reforming without external cracking. Unlike SOFCs, ammonia-fed AEMFC can operate at operating temperatures up to 25–100 $^\circ\text{C}$ and converts the chemical energy in ammonia directly into electrical energy, which has a more compact system. The working principle is that ammonia is oxidized at the anode ($2\text{NH}_3 + 6\text{OH}^- \rightarrow \text{N}_2 + 6\text{H}_2\text{O} + 6\text{e}^-$) in an alkaline electrolyte, while oxygen is reduced at the cathode ($\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$). Electrons flow externally to produce electricity. Although this technology can be applied under ambient conditions, the

Table 1

Overview of five ammonia-to-electricity technologies.

Parameter	Ammonia ICE [31,32]	Ammonia SOFC [30,33]	Direct Low-Temp AFC [34,35]	Cracking-PEMFC [36–38]	Cracking-SOFC [39]
Full-load efficiency	20–35 %	40–60 %	15–30 %	30–50 %	45–65 %
50 % load efficiency	15–25 %	35–55 %	10–20 %	45–50 %	40–60 %
Operating temperature	80–120 $^\circ\text{C}$ (coolant)	600–1000 $^\circ\text{C}$	20–80 $^\circ\text{C}$	60–80 $^\circ\text{C}$ (PEMFC); 400–800 $^\circ\text{C}$ (cracking)	600–1000 $^\circ\text{C}$ (SOFC + cracking)
Fuel requirement	Ammonia (direct)	Ammonia (direct)	Ammonia (direct)	>99.99 % H_2 (from NH_3)	H_2 - N_2 mix (from NH_3 , no strict purity)
Capital cost (\$/kW)	500–1000	2000–4000	800–1500	1500–3500	2500–4500
Lifespan (Years)	10–15	5–10	2–5	3–5 (PEMFC); 10+ (cracking)	5–10 (SOFC); 10+ (cracking)
Emissions	NO_x (0.5–5 g/kWh), trace NH_3	Negligible (N_2 , H_2O)	Trace NH_3 (unreacted)	Negligible (H_2O)	Negligible (N_2 , H_2O)
Startup time	Minutes	Hours (preheating)	Seconds-Minutes	Minutes (cracking preheat)	Hours (preheating)
Scalability	Good (mature supply chain)	Moderate (high-temp challenges)	Excellent (small-scale)	Moderate (PEMFC limits)	Excellent (large-scale)
Key advantages	Low capital cost; mature tech	High efficiency; fuel flexibility	Fast startup; low temp	High part-load efficiency	Highest system efficiency; heat integration
Key disadvantages	Low efficiency; NO_x emissions	High cost; slow startup	Low efficiency; short lifespan	High H_2 purity needs; Pt cost	High cost; slow startup

energy efficiency is limited to 15–30 % due to slow reaction kinetics.

In addition to direct utilization, ammonia can be indirectly applied to fuel cells for electricity generation. This can be realized via ammonia cracking, which allows the production of hydrogen from ammonia, and then the produced hydrogen can be used in PEMFC (with purified) and SOFC (without purified). Taking PEMFC-based system as an example, this hybrid system first cracks ammonia into H_2 and N_2 ($2NH_3 \rightarrow 3H_2 + N_2$), with H_2 fed to a PEMFC for electricity generation. The working principle is that ammonia is cracked at 400–800 °C using a catalyst (e.g., nickel), then purified to >99.99 % H_2 . The H_2 reacts in the PEMFC anode ($H_2 \rightarrow 2H^+ + 2e^-$), with electrons generating current and H^+ combining with O_2 at the cathode to form H_2O . The system efficiency (including cracking) is 30–50 % at full load, with part-load efficiency remaining high (45–50 % at 50 % load) due to stable PEMFC performance. Similar to Cracking-PEMFC, but H_2 from ammonia cracking is fed to a SOFC, leveraging SOFC's tolerance to impure H_2 . The working principle is that ammonia is cracked (same as above), and the H_2 - N_2 mixture (no strict purification needed) is fed to the SOFC anode. H_2 oxidizes at high temperature (600–1000 °C) to produce electricity, with N_2 acting as an inert diluent. The system efficiency is 45–65 % at full load, the highest among the five technologies, due to SOFC's high efficiency and heat integration between cracking (exothermic) and SOFC (endothermic). A comprehensive comparison of the above mentioned five technologies has been summarized in Table 1.

2.2. Economic assessment

2.2.1. Levelized cost

The economic assessment of using ammonia as an energy carrier in this study is calculated via the following method. First of all, using ammonia as an energy carrier includes two connectable processes, i.e., electricity to ammonia and ammonia to electricity, as described in Fig. 1. Following this, the input energy and output energy of each energy conversion unit should be determined in accordance with the energy requirements of the application terminals. In this study, the electricity generated via the proposed ammonia-mediated system is assumed to power a cruise ship. By investigating the energy consumed by the cruise ship, which is around 2710.3 MWh/year (see Table S1), the energy generated from the ammonia applications (such as ICE and fuel cells) can be determined. Next, the total energy input into the ICE and fuel cells can also be determined via the assumed energy conversion efficiencies. Following this, the total energy input and output of each unit, including ammonia storage and transportation, ammonia synthesis, hydrogen production, and renewable electricity generation, can be determined step by step (See Table S3). In addition, according to the heating value of the ammonia and hydrogen, the total amount weight of these chemicals can be determined. Afterward, the size or scale of different units for energy conversion and storage could thus be calculated. Once the technical parameters have been determined, the economic assessment could be carried out via the following equations:

$$IC = \frac{E_t}{CF \cdot \text{Annual hours}} \quad (1.1)$$

Where IC is the installed capacity of the different plants (MW), E_t is the annual electricity generation demand in year t (MWh). CF is the capacity factor of the various plants, referring to the relationship between the factory's actual production capacity and the rated capacity, which is affected by weather, season, factory downtime maintenance time, etc.

$$TC = C_t \cdot IC + O_t \cdot n \quad (1.2)$$

Where TC is the total cost of the project (USD), referring to the actual investment cost of each factory construction, which is related to the capital expenditure in year t (C_t) and operating expense in year t (O_t). n is the project lifetime (in years).

$$LCOE = \frac{\sum_{t=1}^n \frac{C_t + O_t}{(1+r)^t}}{\sum_{t=1}^n \frac{E_t}{(1+r)^t}} \quad (1.3)$$

Where LCOE is the levelized cost of electricity (USD/kWh), which refers to the ratio of the total costs generated by each part of the ammonia value chain to the total output energy. r is the discount rate, if applicable, which is a ratio used to measure the present value of future cash flows.

As demonstrated, the cost considered in this study for each energy conversion unit includes capital expenditure and operating expenses (including maintenance and replacement costs). Following equations (1.1) to (1.3), the total cost of each unit, including renewable electricity generation, hydrogen production, ammonia production, ammonia storage & transportation, and ammonia applications, can be determined. Therefore, the LCOE of each route can be calculated and analyzed.

2.2.2. Sensitivity analysis

The sensitivity analysis in this work includes two parts. First of all, considering the whole energy transition routes, i.e., electricity-ammonia-electricity, there are five different interconnected energy storage and conversion routes. The impact of each unit on efficiency and cost is worth analyzing, which can provide insight into future development. Secondly, for a specific energy storage and conversion unit, a number of components are integrated, which sensitivity should also be investigated. The results of the sensitivity analysis will be introduced in the next section.

3. Results and discussion

3.1. Ammonia production

In accordance with the aforementioned analysis, the electricity-to-ammonia process is achieved via coupling the traditional Haber-Bosch process with water electrolysis. Following this, the cost-affecting parameters, including the capital expenditure (CAPEX) and the operating expense (OPEX, replacement costs are included), for electricity to ammonia, including renewable electricity generation, hydrogen production, as well as green ammonia production, should be investigated respectively.

Regarding renewable electricity, the solar PV (0.049 USD/kW, IRENA 2023) and onshore wind (0.033 USD/kW, IRENA 2023) power plants are selected to generate green electricity due to the cost consideration (see Table S2). In terms of the life span, it is assumed that all power plant equipment is 30 years after reviewing the data from literature and industry. Besides, considering the annual operating hours, it is related to the capacity factors of each type of power plant. The capacity factor here represents the proportion of actual electricity production to the theoretical maximum potential output of a facility. For example, the capacity factor for a solar power production plant is 0.2, and for a wind power production plant is 0.3 due to their intermittent properties [40] (Table 2). In this study, the scale of the renewable power plant is assumed to be 1 MW or 100 MW in accordance with the estimated energy requirements for terminal applications. Considering the scale effect on cost, for example, the CAPEX of the solar power plants of 1 MW and 100 MW is calculated to be 748 USD/kW and 506 USD/kW, respectively [41]. The scale of wind power plants contains two types, onshore and offshore wind power plants, the average CAPEX is 1250 USD/kW [42]. Besides, the OPEX of both solar and wind power plants is 2 % of the CAPEX per year. To calculate the cost of the renewable power, the size should be determined first. Take the ICE as the terminal application as an example, the ICE-based electricity to ammonia refers to the parts of the ammonia value chain containing renewable electricity generation, green hydrogen production, and green ammonia production. The specific electrical requirement from the power plant (see Table S3) is 13355.97 MWh/year. According to the capacity factor, the installed

Table 2

The techno-economic characteristics of the process of electricity to ammonia.

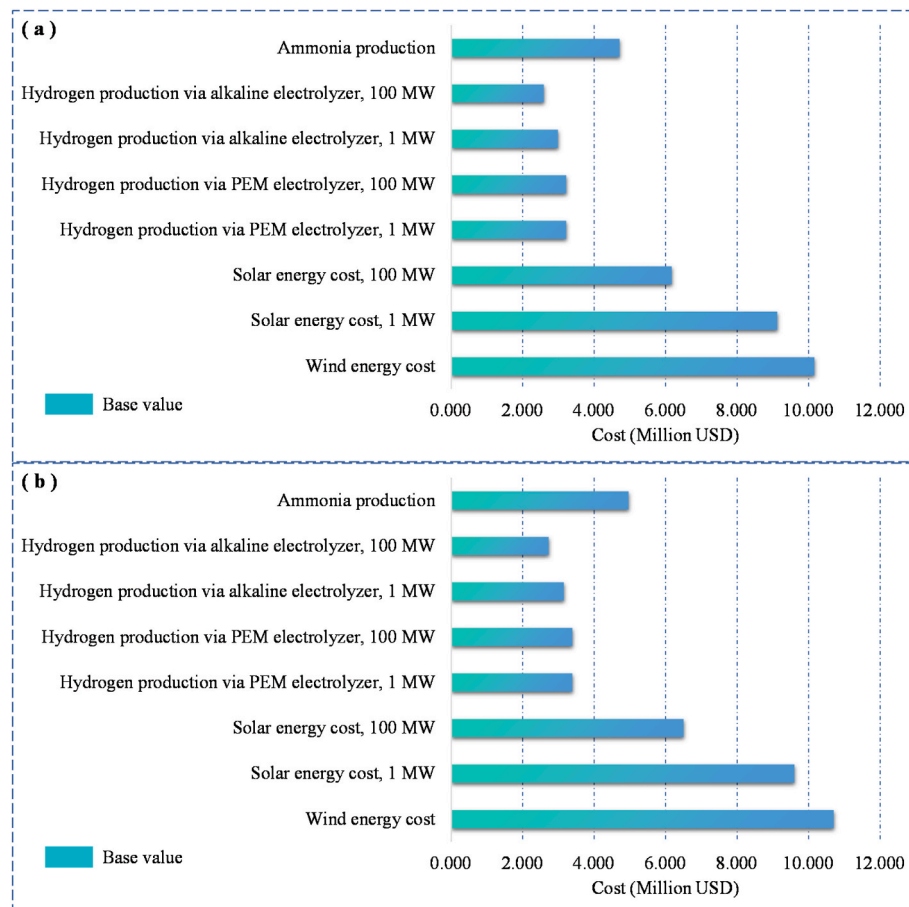
Stages	Capacity factor	Energy efficiency (%)	CAPEX (USD/kW)	OPEX (USD/kW)	LCOE (USD/kWh)	Ref.
Wind energy	0.3	20	1250	2 % of CAPEX	0.02536	[42]
Solar energy (1 MW)	0.2	20	748.0	2 % of CAPEX	0.02276	[41]
Solar energy (50 MW)	0.2	20	635.8	2 % of CAPEX	0.01934	[46]
Solar energy (100 MW)	0.2	20	506.0	2 % of CAPEX	0.01540	[47]
PEM electrolyzer (1 MW)	0.8	72	1017.5	5 % of CAPEX	0.01208	[48]
PEM electrolyzer (100 MW)	0.8	72	1017.5	5 % of CAPEX	0.01208	[48]
Alkaline electrolyzer (1 MW)	0.7	66.5	825	5 % of CAPEX	0.01120	[48]
Alkaline electrolyzer (100 MW)	0.7	66.5	715	5 % of CAPEX	0.00972	[48]
Ammonia production	0.8	66.5	1870	5 % of CAPEX	0.02224	[49]

capacity of solar and wind power production plants is calculated to be 7.62 MW and 5.023 MW respectively. Therefore, the total cost of the solar power production plants is 9.12 million USD (1 MW) and 6.17 million USD (100 MW). The wind power production plants onshore and offshore are 10.16 million USD (Fig. 2).

However, when the fuel cells were applied to generate electricity for the terminal application, the electricity requirements could be different. Taking the SOFC as an example, it only requires 9713.44 MWh due to higher energy conversion efficiency, resulting in the installed capacity of solar and wind power production plants being reduced to 5.54 MW and 3.70 MW, respectively. Therefore, the total calculated cost of the solar power production plants is 6.64 million USD (1 MW) and 4.49 million USD (100 MW). The wind power production plant is 7.39 million USD. In comparison, the AEMFC has the lowest energy efficiency (35 %), resulting in the highest electricity requirement of 15263.83 MWh. Therefore, the total cost of renewable electricity from the solar power production plants is 10.41 million USD (1 MW), and 7.04 million USD

(100 MW), respectively, while for the wind power production plant is 11.60 million USD. Therefore, following equation (1.1) and equation (1.3), the lowest levelized cost of electricity (LCOE) of renewable electricity generated from solar plants (100 MW) is 0.0154 USD/kWh in 2023, which is suitable for all terminal applications.

Following the economic analysis of renewable electricity, the cost of hydrogen production will be investigated. Regarding the maturity of the technology as well as the achievable cost of hydrogen production, the AWE and PEMWE technologies are selected to generate green hydrogen. Considering the steady-state operating time as well as the operations & maintenance time, the capacity factor of the AWE plants is 0.7 [43], and the PEMWE plants is 0.8 [44] (Table 2). In this work, the scale of hydrogen production plants is assumed to be 1 MW or 100 MW in accordance with the estimated energy requirements for terminal applications. For example, the CAPEX of the alkaline water electrolysis plants is 815 USD/kW (1 MW), and 715 USD/kW (100 MW). In addition, The CAPEX of the PEMWE plants of 1 MW and 100 MW both are 1017.5

**Fig. 2.** (a) Cost analysis of ICE-based electricity to ammonia; (b) Cost analysis of PEMFC-based electricity to ammonia.

USD/kW due to the lower technological maturity, which will be less affected by the economies of scale [45]. In terms of the OPEX of both AWE and PEMWE plants, which is assumed to be 5 % of the CAPEX per year [40]. To calculate the total cost, the hydrogen production size should be determined. Taking the ICE as an example, the total energy contained in the produced hydrogen is 8881.70 MWh. According to the capacity factor, the installed capacity of alkaline water electrolysis plants and PEMWE plants are 1.45 MW and 1.27 MW respectively. Therefore, the total cost of the AWE plants is 2.99 million USD (1 MW) and 2.59 million USD (100 MW). While for the PEMWE plants, the total cost is 3.22 million USD (1 MW & 100 MW) (Fig. 2). Following these, the lowest production cost of green hydrogen is the use of 100 MW AWE technology, with a levelized cost of hydrogen (LCOH) of 0.0097 USD/kWh in 2023. In terms of the investment cost, it is expected to drop by 61.4 % to 1.00 million USD by 2050 due to the technology development.

Regarding the ammonia production via the Haber-Bosch process, the capacity factor of the ammonia production plants is 0.8 (Table 2). The CAPEX of the H-B process ammonia synthesis plants is 1.87 million USD/MW. The OPEX is 5 % of the CAPEX per year. Similarly, take the ICE as an example, the energy requirement which is needed to be output by the ammonia production plants is 7058.09 MWh, which means that 1357.35 tons of ammonia needs to be synthesized. According to the capacity factor, the installed capacity of ammonia production plants is 1.01 MW. Therefore, the total cost of ammonia synthesis is 4.71 million USD (Fig. 2). Besides, the SOFC-based, AEMFC-based, and SOFC (cracker)-based electricity-ammonia cost analysis are illustrated in Figure S1-S3.

To eliminate the effect of the energy conversion rate of the terminal application, using the levelized cost of ammonia (LCOA) to measure economics. Following these, the LCOA in 2023 is 0.0224 USD/kWh. In terms of the investment cost, it is expected to drop by 47.1 % to 2.49 million USD by 2050.

In summary, the pathway of using a 100 MW solar power plant to generate electricity, powering 100 MW alkaline water electrolyzers (AWE) for green hydrogen production, and subsequently synthesizing green ammonia, achieves a total investment cost of \$13.47 million and a levelized cost of \$0.0474/kWh in 2023, with projections to drop to \$0.0232/kWh. This signals that renewable energy-based green ammonia production is becoming economically viable, offering a cost-competitive solution for energy storage and conversion. To further reduce the operational cost, future development perspectives are summarized as follows: 1) Renewable electricity generation (Solar/Wind): to minimize costs in electricity generation, advancements in solar and wind technologies are essential. High-efficiency photovoltaic systems, such as tandem solar cells (e.g., perovskite-silicon), can achieve efficiencies exceeding 30 %, while larger and more efficient wind turbines (15+ MW offshore models) reduce levelized costs of energy (LCOE). Hybrid renewable systems—combining solar, wind, and battery storage—help mitigate intermittency and improve capacity factors. Additionally, AI-based energy management systems can optimize power dispatch, and advanced tracking (e.g., bifacial panels with single-axis tracking) can increase solar yields by 10–25 %. 2) Hydrogen production via water electrolysis (AWE/PEMWE): Electrolyzer efficiency and cost are critical determinants of green ammonia's viability. Alkaline water electrolyzers (AWE) benefit from improved electrode materials (e.g., Ni-Fe coatings) that reduce overpotential losses, while proton exchange membrane electrolyzers (PEMWE) could adopt non-precious metal catalysts (e.g., Fe/Ni-based) to lower costs. High-temperature solid oxide electrolysis (SOEC) presents another promising avenue, particularly when integrated with waste heat from ammonia synthesis, achieving system efficiencies above 85 %. Economically, mass manufacturing of electrolyzers at gigawatt scale—as seen in China's recent expansions—can substantially reduce unit costs. Additionally, securing low-cost renewable electricity through off-peak power contracts and co-locating electrolyzers with ammonia plants minimizes transportation

expenses. On the policy front, carbon pricing on grey hydrogen (produced via steam methane reforming) and government-backed procurement programs for green hydrogen can stimulate market demand and investment. 3) Ammonia synthesis via the Haber-Bosch process: the Haber-Bosch process remains energy-intensive, but innovations in catalyst technology and reactor design can improve efficiency. Ruthenium-based catalysts enable lower-pressure operation, reducing energy consumption, while modular small-scale reactors enhance flexibility for decentralized production. Emerging alternatives, such as plasma-driven and electrochemical ammonia synthesis, could bypass traditional Haber-Bosch constraints entirely, though further research is needed to scale these methods. Economically, integrating ammonia synthesis with hydrogen production facilities eliminates transportation costs, and carbon credit mechanisms can improve financial viability. Policy measures, such as mandates for green ammonia blending in fertilizers and shipping fuels, along with subsidies for export-oriented green ammonia corridors, can create stable demand and incentivize industry adoption.

Overall, achieving cost-competitive green ammonia hinges on a multi-faceted approach. Technologically, advancements in renewable energy, electrolysis, and synthesis methods must be pursued. Economically, scaling production, optimizing supply chains, and securing financing are crucial. Policy frameworks—including carbon pricing, R&D incentives, and market mandates—play an indispensable role in accelerating adoption. By addressing these dimensions holistically, green ammonia can emerge as a cornerstone of the global energy transition, enabling decarbonization in sectors such as shipping, power generation, and heavy industry.

3.2. Electricity generation

The ammonia to electricity process mainly involves ammonia storage, transportation, and terminal applications. Economic parameters considered during the above process consist of the CAPEX and the OPEX, (replacement costs are included as well). Therefore, to obtain the economic data on electricity generation, cost parameters of various stages should be investigated. Due to the fact that ammonia has been used as a principal component in chemical feedstocks and fertilizers for many decades, the infrastructure for ammonia transportation, and storage has been well established. In terms of transportation, ammonia can be conveyed through multiple ways, including trucks, pipelines, railways, and maritime shipping. Among these technologies, maritime transport is deemed more advantageous for long-distance storage and transportation [50]. In this work, ammonia transportation focuses on investigating the economic impacts of maritime transport, with all economic data being predicated on the assumption of a shipping distance of 1500 km [51]. Considering the liquid ammonia can be stored at room temperature, and the design pressure of the ammonia storage tank is ~11–15 bar. To coincide with the equipment lifespan of ammonia production facilities, the operational period for the storage and transportation of ammonia is assumed to be 30 years. It is worth mentioning that the cost of ammonia transport and storage is independent of annual operating hours and capacity factor, which is solely related to the mass of ammonia. In this work, the mass of liquid ammonia is represented by its contained energy. Generally, 1 kg of ammonia can supply the energy of 18.6 MJ by the ammonia combustion, which means it contains an energy of 5.17 kWh/kg. For example, the CAPEX of the ammonia transport is 0.0364 USD/kWh and the ammonia storage is 0.0091 USD/kWh in 2023 [51]. The OPEX of both of them is 2 % of the CAPEX per year. Take the ICE as the terminal application as an example, the ICE-based ammonia to electricity refers to the parts of the ammonia value chain containing ammonia transportation, ammonia storage, and clean electricity generation. The energy output from the ammonia production facility is 7058.09 MWh. Accounting for a 4 % energy loss of the transportation and storage process, the energy content of ammonia requiring transport and storage is calculated to be 6775.75 MWh. Therefore, the total cost of

ammonia transport by ship is 0.39 million USD, and the total cost of ammonia storage is 0.10 million USD, respectively. The ammonia storage process refers to the ammonia being stored at the departure and arrival ports of maritime transportation. Similarly, to eliminate the effect of the energy conversion rate of the terminal application, the levelized cost of transport (LCOT) and levelized cost of storage (LCOS) are applied to measure the economics. Following these, the LCOT of green ammonia is 0.0019 USD/kWh and the LCOS of green ammonia is 0.0005 USD/kWh in 2023. In terms of the investment cost, due to the technology development, the cost of ammonia transport is expected to drop by 38.5 % to 0.24 million USD by 2050 and the cost of storage is

expected to drop by 60.0 % to 0.04 million USD by 2050, respectively.

Regarding the maturity and energy conversion rate of the ammonia application terminal, the five cases are investigated for utilizing the ammonia to generate electricity. The ICE can directly burn ammonia as fuel converting the chemical energy into thermal energy, which is cleaner than traditional fuels. Fuel cells, which can directly convert the chemical energy stored in the ammonia into electrical energy, are considered as one of the ideal ways to effectively utilize ammonia.

After reviewing the data from the literature and industry, the CAPEX associated with ICE technology stands at 521 USD/kW, showcasing its relative affordability in the current market [52]. In contrast, Solid Oxide

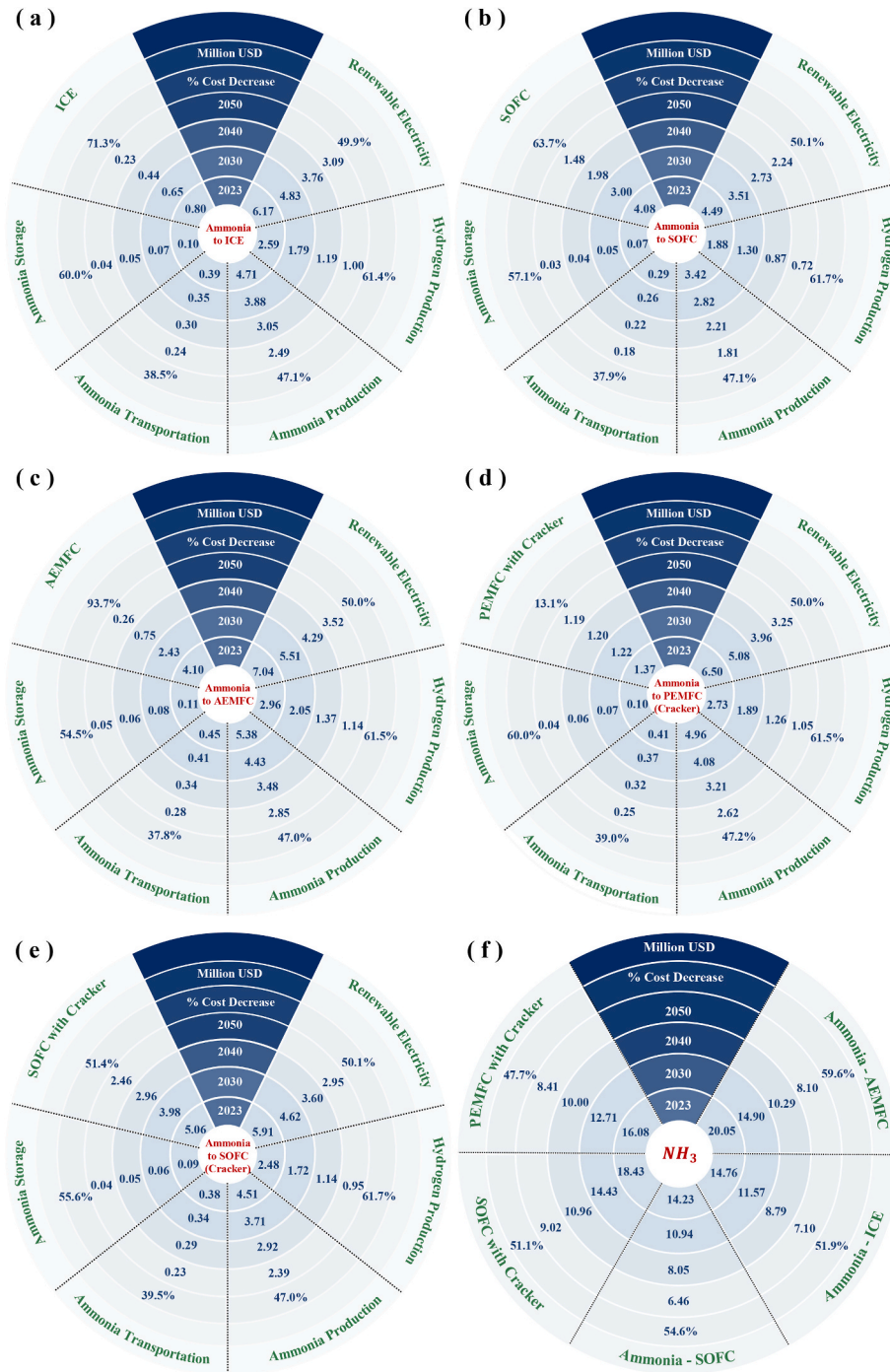


Fig. 3. Cost analysis of energy conversion systems at different time periods (2023, 2030, 2040, 2050) (a) ICE-based electricity-ammonia-electricity route; (b) SOFC-based electricity-ammonia-electricity route; (c) AEMFC-based electricity-ammonia-electricity route; (d) SOFC-based electricity-ammonia-hydrogen-electricity route; (e) PEMFC-based electricity-ammonia-hydrogen-electricity route; (f) Cost comparison of five different routes.

Fuel Cell (SOFC) systems exhibit a higher CAPEX of 2200 USD/kW, reflecting their advanced technology but also indicating higher initial investment costs [53]. Similarly, Anion Exchange Membrane Fuel Cells (AEMFCs) are characterized by a CAPEX of 2212 USD/kW, aligning them closely with SOFCs in terms of financial outlay [53]. Proton Exchange Membrane Fuel Cells (PEMFCs), however, present a considerably lower CAPEX of 166 USD/kW, suggesting a more economically viable option among other fuel cell technologies [54]. Additionally, the indirect utilization of ammonia necessitates the inclusion of an ammonia cracking unit, which bears a CAPEX of 153 USD/kWh [55]. In this work, to match the electricity requirements of the application terminals, all ammonia-utilizing fuel cell systems are assumed to have a capacity of 700 kW. Therefore, the findings reveal that the total investment cost for the ICE system stands at 0.80 million USD with a levelized cost of electricity (LCOE) of 0.0098 USD/kWh. The SOFC system exhibits a higher investment cost of 4.08 million USD and an LCOE of 0.0502 USD/kWh. Similarly, the AEMFC system's total cost is reported at 4.10 million USD, with its LCOE calculated to be 0.0504 USD/kWh. The PEMFC system with a cracker results in a total cost of 1.37 million USD and an LCOE of 0.0102 USD/kWh, while the SOFC system with a cracker incurs a total cost of 5.06 million USD and an LCOE of 0.0568 USD/kWh (Fig. 3). The detailed economic characteristics of ICE and fuel cells are illustrated in Table 3. Furthermore, all of these technologies showed a significant reduction in the investment cost for the ammonia-electricity system during the time period of 2023–2050. For example, the ICE systems anticipate a decrease of 71.3 % to \$0.23 million by the year 2050. From the perspective of the entire ammonia value chain, the total cost of the ICE application route is 14.76 million USD in 2023, containing the ammonia production and electricity generation. Additionally, the total cost is expected to drop by 51.9 % to 7.10 million USD by 2050. In summary, the transition to ammonia as a carbon-free energy carrier necessitates substantial improvements in the efficiency and cost-competitiveness of its conversion to electricity. For ammonia-fueled internal combustion engines (ICEs), advancements in turbulent jet ignition systems, optimal ammonia-hydrogen blends, and waste heat recovery can significantly enhance combustion efficiency and power output, while policy incentives for early adopters in maritime and stationary power applications could accelerate deployment. Solid oxide fuel cells (SOFCs) operating directly on ammonia require improved thermal integration between cracking and electrochemical processes, durable anode materials resistant to nitriding, and standardized manufacturing to reduce costs, supported by government-funded demonstration projects in shipping and industrial power. Direct low-temperature ammonia fuel cells (AEM-DAFCs) face challenges in membrane stability and catalyst costs, necessitating research into alkaline anion exchange membranes (AEMs) and non-precious metal

catalysts, alongside policy inclusion in renewable energy incentives. Ammonia cracking for PEMFCs demands innovations in low-energy cracking reactors and purification systems to minimize hydrogen production costs, while regulatory frameworks for “green hydrogen” derived from ammonia could enhance economic feasibility. Similarly, ammonia cracking for SOFCs benefits from internal cracking integration and hybrid system designs, with policy support needed to recognize their efficiency advantages over standalone PEMFC solutions. Cross-cutting strategies include standardized ammonia quality specifications, carbon pricing mechanisms, and AI-driven optimization of integrated ammonia production and power generation systems. Collectively, these technical, economic, and policy measures can position ammonia as a scalable and efficient zero-carbon energy vector, enabling its widespread adoption in power generation, maritime transport, and industrial applications.

3.3. System efficiency and scale impacts

The efficiency of energy conversion systems in the ammonia value chain is a critical determinant in assessing the feasibility of electricity-ammonia-electricity for large-scale deployment. In this work, the initial electricity was assumed to be directly generated from solar energy and wind energy, therefore the efficiency of renewable sources to renewable electricity was not considered. At the electrolyzer stage, the PEMWE operates at an efficiency of 72 %, while the AWE works at a slightly lower efficiency of ~66.5 %. The efficiency mentioned above means the percentage of electrical energy contained in the electricity to the chemical energy stored in the produced hydrogen. In the ammonia synthesis stage, the improved Haber-Bosch process, supplemented by air separation to extract nitrogen, attains an efficiency of ~98 % through the implementation of steam cycle optimization and integration technology.

For ammonia application terminals, the energy efficiency of ammonia transportation and storage is also considerable, calculated at ~96 %. The main energy consumption is attributed to the power consumption of compressors and other auxiliary machinery, while there is a small amount of ammonia loss. Regarding the energy conversion technologies for electricity generation, notable variations are observed in the energy conversion efficiency. When the ammonia was used directly as fuel in the ICE, a ~40 % conversion efficiency was achieved. In contrast, when liquid ammonia is utilized to supply energy to AEMFC, a larger input of 7743.71 MWh is required to meet an output demand of 2710.3 MWh, resulting in an efficiency of only ~35 %. For SOFC, the efficiency improves to ~55 %, which requires an input of 4927.82 MWh for the same output, attributable to the cogeneration capacity of heat and power in SOFC. In scenarios where liquid ammonia is indirectly employed, i.e., ammonia was converted to hydrogen through a cracker before powering PEMFC and SOFC, the cracker efficiency ranges from ~70 to 79 %. Additionally, when hydrogen is utilized to supply energy to PEMFC, a middle input of 5420.60 MWh is required to meet an output demand of 2710.3 MWh, which results in an efficiency of ~50 %.

An illustration of the energy conversion efficiencies from 2023 to 2050 of the whole ammonia value chain, which includes the renewable electricity to ammonia to clean electricity generated by five different technologies, was presented in Figs. 4 and 5. For instance, the conversion efficiency from renewable electricity to ammonia to clean electricity generated by AEMFC is anticipated to rise from 21.02 % to 29.59 % by 2050. This progression points to advancements in research and development, including improved catalyst design, system integration, and material science, which enhance the performance of AEMFC. The overarching trend reflects a positive outlook on ammonia's evolving role as a sustainable and efficient energy carrier, especially as enhancements in efficiency are projected to continue.

The scale of production facilities significantly influences the levelized cost of generating renewable electricity and green hydrogen, integral components in the ammonia value chain. This influence is

Table 3
The economic characteristics of the ICE and fuel cells.

Application terminals	Power (kW)	CAPEX (USD/kW)	O & M cost (USD/kW)	Total cost (Million USD)	LCOE (USD/kWh)	Ref.
ICE	700	521	4 % of the CAPEX	0.80	0.00984	[52]
SOFC	700	2200	5.5 % of the CAPEX	4.08	0.05018	[53]
AEMFC	700	2212	5.5 % of the CAPEX	4.10	0.05042	[56]
PEMFC-cracker	700	163	5 % of the CAPEX	1.37	0.01021	[54]
SOFC-cracker	700	2200	5.5 % of the CAPEX	5.06	0.05682	[53]

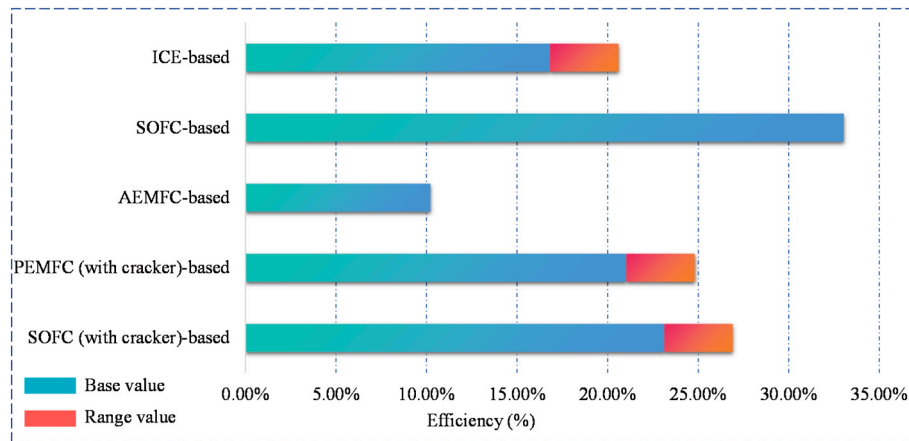


Fig. 4. Energy efficiency comparison at five electricity-ammonia-electricity routes in 2023.

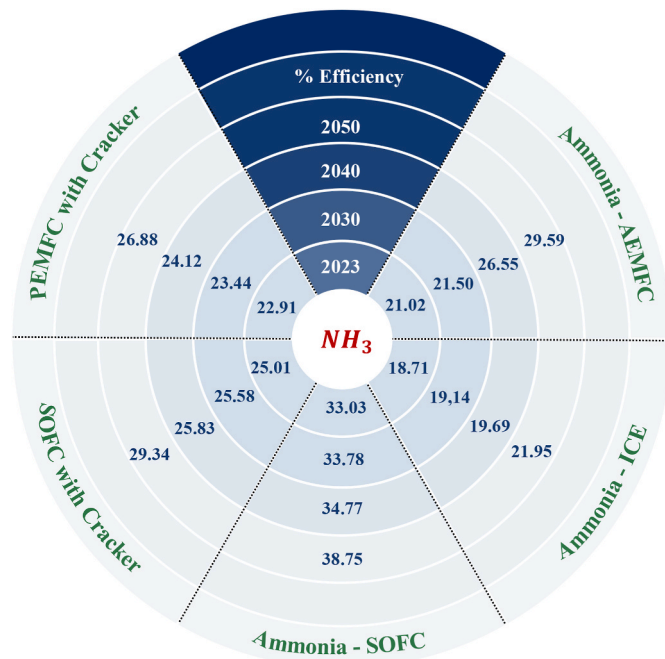


Fig. 5. Energy efficiency comparison at different time periods (2023, 2030, 2040, 2050) of five electricity-ammonia-electricity routes.

primarily attributed to economies of scale, where an increase in plant size corresponds to a reduction in both the installation and production costs per unit, hence lowering the CAPEX. Meanwhile, a larger scale also leads to the OPEX reduced due to lower unit operation and maintenance costs, which are substantial components of a plant's life-cycle production expenses. Therefore, the cheapest cost the renewable electricity is generated by the solar power plants of 100 MW, and the cheapest green hydrogen produced by the AWE hydrogen production plants of 100 MW (Table 4). For instance, a solar power plant scaled to 100 MW exist a CAPEX of 506 USD/kW, decreased by 242 USD/kW compared with the solar power plant of 1 MW. Meanwhile, an AWE hydrogen production plant of 100 MW exist a CAPEX of 715 USD/kW, decreased by 110 USD/kW compared with the plants of 1 MW. The forecast of the ammonia value chain is carried out in the ten-year phase from 2030 to 2050 considering the technological advancements within the context of economic viability. Considering the ammonia direct utilization, the decreasing trend in the levelized cost of electricity (LCOE) for the ammonia-to-ICE could be illustrated, with the reduction in LCOE from 0.068 USD/kWh to 0.032 USD/kWh for plant sizes of 1 MW from 2023

Table 4

The scale effect on the system cost at different time periods (2023–2050).

Years	2023	2030	2040	2050	Ref.
Electricity generation via solar energy					
Solar cost (USD/kW) industrial, 1 MW	748.0	583.0	440.0	363.0	[41]
Solar cost (USD/kW) Utility-scale, 50 MW	635.8	495.6	374.0	308.6	[46]
Solar cost (USD/kW) Utility-scale, 100 MW	506.0	396.0	308.6	253.0	[57]
Hydrogen production via PEM electrolyzer					
Cost (USD/kW) 1 MW	1017.5	715	495	440	[48]
Cost (USD/kW) 100 MW	1017.5	715	495	440	[48]
Hydrogen production via alkaline electrolyzer					
Cost (USD/kW) 1 MW	825	627	495	385	[48]
Cost (USD/kW) 100 MW	715	495	330	275	[48]

to 2050 (Fig. S4). The decreasing trend in the levelized cost of LCOE for the ammonia-to-SOFC could be illustrated, with the reduction in LCOE from 0.109 USD/kWh to 0.048 USD/kWh for plant sizes of 1 MW from 2023 to 2050 (Fig. 6). While for the ammonia-to-AEMFC route, a reduction in LCOE from 0.109 USD/kWh to 0.033 USD/kWh was achieved from 2023 to 2050 (Fig. 6). Similarly, the ammonia indirect utilization, the decreasing trend in the LCOE for the ammonia-to-PEMFC route could be illustrated, with the reduction in LCOE from 0.069 USD/kWh to 0.037 USD/kWh for plants size of 1 MW, spanning from 2023 to 2050 (Fig. 7). Regarding the ammonia-to-SOFC value route, the LCOE decreases from US \$0.116/kWh to US \$0.054/kWh from 2023 to 2050 (Fig. 7). At present, the LCOE within the ammonia value chain is predominantly dictated by the investment costs in fuel cell technology, owing to the necessity for precious metal catalysts and intricate manufacturing processes. However, with the anticipated progress in science and technology, the expenditure share of AEMFCs is projected to drop significantly, from 46 % to 10 % within a 1 MW ammonia supply chain, showing enhanced energy efficiency and cost-effectiveness of fuel cells.

In contrast, the cost proportion attributable to ammonia production illustrates a trend of rise, given that the Haber-Bosch process has matured with an efficiency peak at 98 %, leaving limited scope for cost reductions. Moreover, within the ammonia value chain, components such as renewable electricity generation and green hydrogen production display a consistent proportion within the overall LCOE. Notably, the costs associated with ammonia transportation and storage are found to be minimal, only 2 % and 1 %, respectively, and are projected to become negligible by the years 2040 and 2050, underscoring ammonia's competency for long-term energy storage and long-distance transportation solutions. The projected LCOE for each segment of the ammonia supply

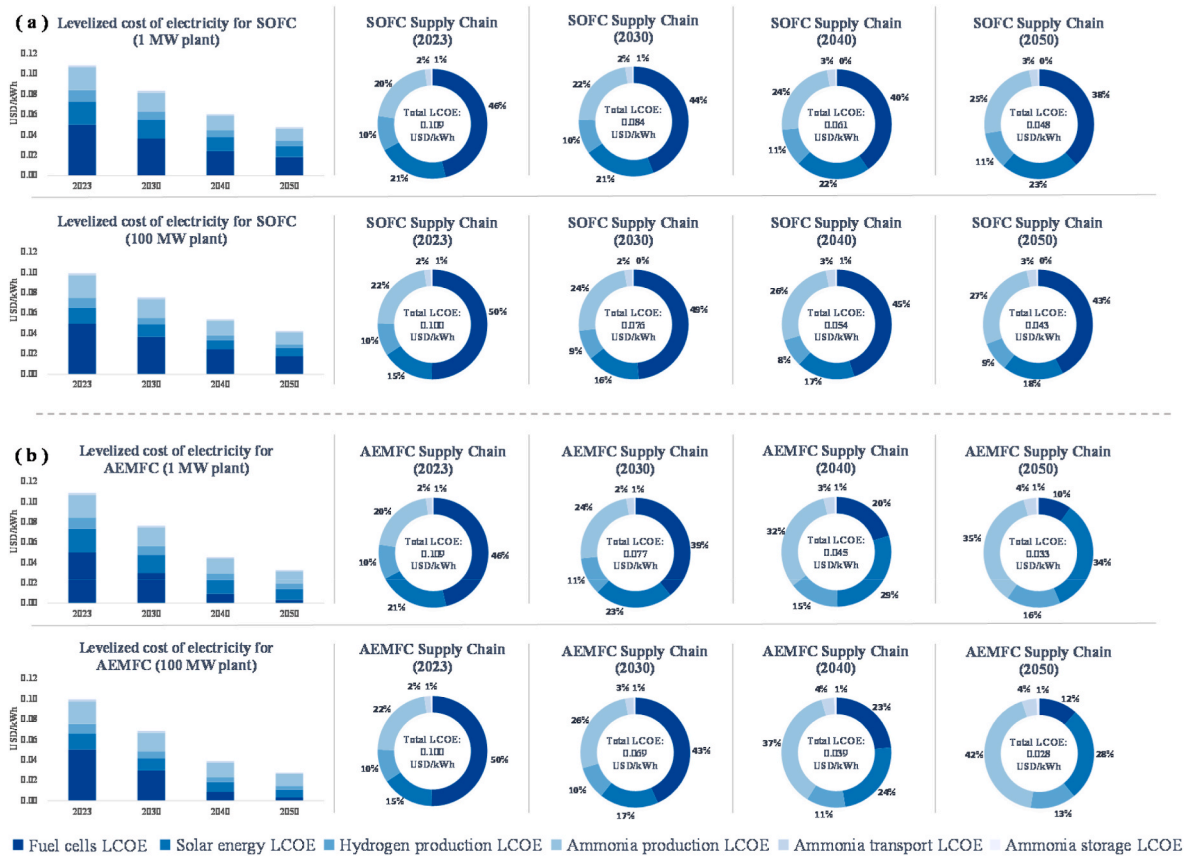


Fig. 6. (a) Size effect on the system cost of SOFC-based electricity-ammonia-electricity route (1 MW & 100 MW); (b) Size effect on the system cost of AEMFC-based electricity-ammonia-electricity route (1 MW & 100 MW).

chain, encompassing renewable electricity, hydrogen production, ammonia synthesis, and the subsequent storage and transport stages, and the ammonia application is on a declining trajectory. Therefore, the application of ammonia as an energy carrier stands to gain from these cost efficiencies, particularly in larger 100 MW plants that benefit from scale economies. In summary, the period between 2030 and 2050 is poised to be a transformative era for the ammonia energy landscape, guided by technological advancements, economic motivations, and environmental imperatives, positioning ammonia as a pivotal player towards a renewable energy paradigm.

To further improve the energy efficiency of the electricity-ammonia-electricity route, various strategies should be carried out via the following two perspectives: 1) Enhancing ammonia production efficiency: To improve the efficiency of green ammonia production, integrating AI-driven renewable energy management can optimize Haber-Bosch operations under variable power inputs. Advanced electrolyzers, particularly PEMWE, should be further developed for flexible operation with intermittent renewables. Process innovations, such as plasma-assisted ammonia synthesis at lower pressures (50 bar vs. traditional 100–400 bar), can reduce energy consumption. Economically, scaling production to 1.5 million tonnes/year is projected to achieve cost parity with fossil-based ammonia by 2028. Policy support through carbon pricing ($\geq \$80/\text{ton CO}_2$) and renewable ammonia certification schemes can accelerate adoption. 2) Optimizing ammonia-to-electricity conversion. For ammonia-fueled ICEs, turbulent jet ignition and waste heat recovery can boost efficiency, while retrofit kits lower transition costs. Direct ammonia SOFCs require durable nitriding-resistant anodes and thermal integration to achieve $>80\%$ system efficiency. Low-temperature AEM-DAFCs need improved membranes and non-precious metal catalysts to reduce costs. Ammonia cracking for PEMFCs/SOFCs benefits from microwave plasma cracking and

integrated purification to minimize energy losses. Standardized manufacturing and modular designs (e.g., AFC Energy's scalable cracking units) can drive down costs across all pathways.

4. Concluding remarks

This study presents a rigorous techno-economic assessment of ammonia as a renewable energy vector within a comprehensive electricity-ammonia-electricity energy storage paradigm. Our investigation provides three principal contributions to the field: (1) a systematic comparative analysis of multiple ammonia-to-power conversion pathways under standardized evaluation criteria, (2) longitudinal cost projections extending to 2050 with technology learning curves, and (3) quantification of scale effects across an order-of-magnitude capacity range (1–100 MW). Key findings from the TEA demonstrate that the electricity-ammonia-electricity route achieves an energy efficiency range of 18.71–33.03 % in 2023, with significant room for improvement: projections indicate efficiency gains of 17.3–40.7 % by 2050. Economically, the PEMFC-based pathway (via ammonia cracking) emerges as the most viable today, with the lowest operational cost (~ 0.06 USD/kWh). Notably, the AEMFC pathway exhibits the strongest cost-reduction potential, with a projected $\sim 59.60\%$ drop by 2050, while scaling system size from 1 MW to 100 MW reduces overall costs by $\sim 10\%$, underscoring the importance of economies of scale.

Critical comparisons between pathways reveal clear trade-offs: ICE-based systems, while established, underperform relative to fuel cell pathways in both energy efficiency and long-term cost reduction, making them less competitive for large-scale renewable energy storage. Among fuel cells, PEMFC (via cracking) balances current economic viability with moderate efficiency, whereas SOFC pathways show higher baseline efficiency but lower near-term cost competitiveness. AEMFC,

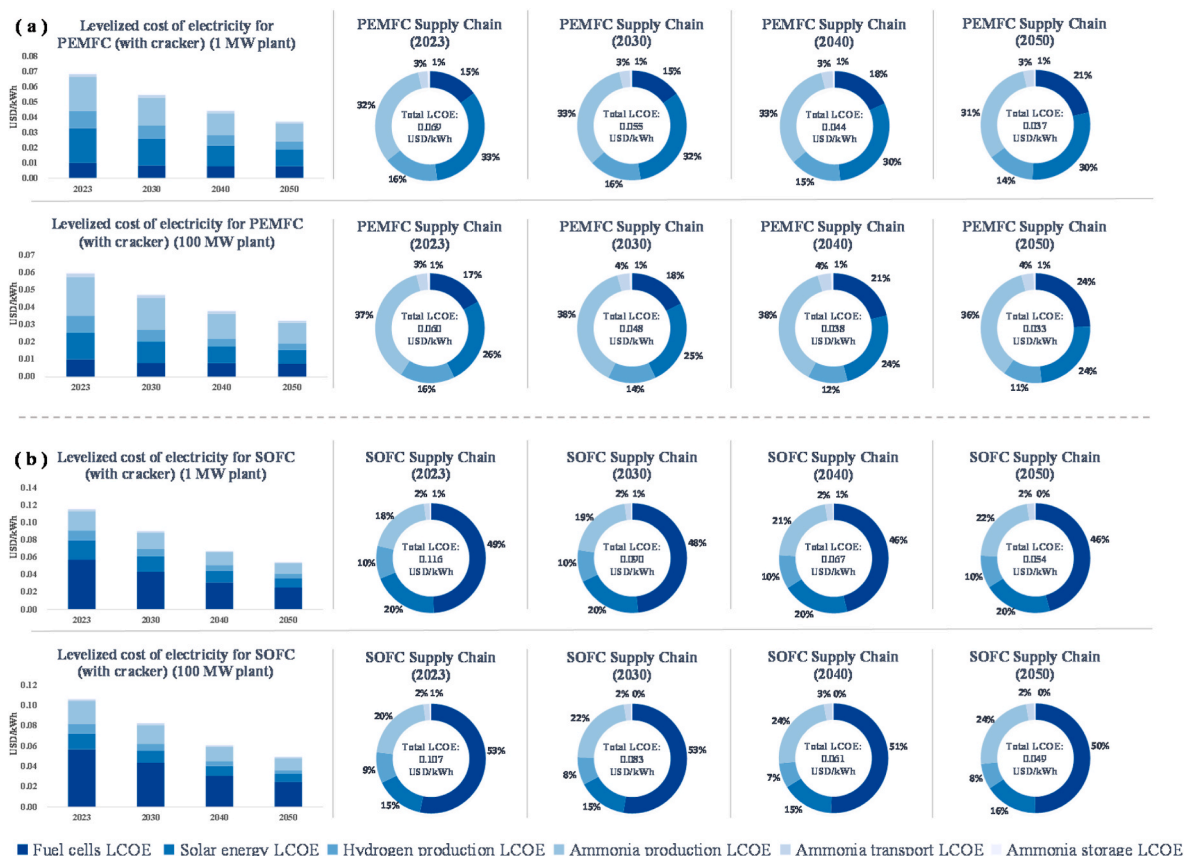


Fig. 7. (a) Size effect on the system cost of PEMFC-based electricity-ammonia-hydrogen-electricity route (1 MW & 100 MW); (b) Size effect on the system cost of SOFC-based electricity-ammonia-hydrogen-electricity route (1 MW & 100 MW).

though currently less economical, offers the most aggressive efficiency and cost improvement trends, positioning it as a strong future contender. These findings inform strategic decisions in energy infrastructure development, supporting the transition to decarbonized systems by identifying high-priority pathways for investment and R&D.

Future efforts to enhance the system should focus on three fronts: (1) improving electrolysis and ammonia synthesis efficiency (e.g., advanced catalysts for Haber-Bosch and water splitting), (2) reducing parasitic losses in ammonia cracking and fuel cell operation via material innovation (e.g., durable AEMFC membranes, low-temperature cracking catalysts), and (3) optimizing system integration to minimize energy losses across the electricity-ammonia-electricity chain. Additionally, scaling production capacities and standardizing components will further drive down costs, while advancing durability testing of fuel cell and ICE components will ensure long-term system stability.

CRediT authorship contribution statement

Yun Liu: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Mingyang Luo:** Writing – original draft, Formal analysis, Data curation. **Wenzhi Li:** Formal analysis, Data curation. **Chuncheng Zhou:** Data curation. **Xiaoyu Huo:** Formal analysis. **Zhefei Pan:** Writing – review & editing, Conceptualization. **Rong Chen:** Conceptualization. **Liang An:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2025.150784>.

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